

ArchaeoGPT: Multi-Step Visual Reasoning for Fine-Grained Painted Pottery Recognition

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Abstract

For over a decade, fine-grained visual classification has been built on an assumption: that subordinate categories are distinguished by localized discriminative features—the bird’s beak, the car’s headlight, the aircraft’s tail fin. But what happens when no single feature is diagnostic, and classification requires composing multiple interacting cues—decorative motifs, vessel morphology, clay texture—each of which constrains the interpretation of the others? We study this question through ancient painted pottery recognition, a task that exposes fundamental limitations of the single-pass feature-matching paradigm. On our newly constructed PotteryFGVC benchmark (2,466 artifacts, 79 cultures, 50 types, 7 eras), we document three findings: (1) CLIP achieves 0.0% zero-shot accuracy, collapsing to predict only 21 of 58 classes—web-scale pre-training provides no prior for domain-specific terminology; (2) naive hierarchical prediction is actively harmful, underperforming flat classification by 20% because hard decisions propagate errors irrecoverably; (3) compositional cues require multi-step integration, with each reasoning step and each visual scale contributing independently to accuracy. These findings motivate ArchaeoGPT, a framework built on a single insight: the model should iteratively refine its interpretation through a continuous hidden state, attending to different visual aspects at each reasoning step, rather than committing to premature discrete decisions. ArchaeoGPT achieves state-of-the-art performance (66.8% on 58-way culture classification, 76.4% on 50-way type classification), with reasoning trajectories showing progressive refinement toward correct predictions. By reframing fine-grained recognition as a compositional reasoning problem, our work opens a direction for domains where categories are defined not by isolated features but by their interaction.

1 Introduction

What makes fine-grained visual classification fine-grained? For over a decade, the answer has been implicit in every benchmark and method design: subordinate categories are distinguished by localized discriminative features. The bird’s beak curvature [?]. The car’s headlight shape [?]. The aircraft’s engine nacelle [?]. This assumption—that somewhere in the image there exists a diagnostic region whose detection suffices for classification—unifies three generations of FGVC methods: part-based detectors that localize semantic regions [?, ?, ?], attention mechanisms that amplify the most discriminative activations [?, ?, ?], and transformer architectures that learn to attend to informative patches [?, ?].

This paper is about what happens when that assumption breaks. Not partially—fundamentally.

Ancient painted pottery presents a recognition problem where no single feature is diagnostic. Consider the task faced by an archaeologist examining a painted sherd excavated from the upper Yellow River valley. The vessel’s outward-flared rim and elongated neck suggest a water-pouring vessel. The red-painted spiral-wave motif on the upper body is characteristic of the Majiayao ceramic tradition (3100–2700 BCE). But the black-banded rim treatment points specifically to the Machang sub-type, while the fine-textured orange clay body suggests late-phase production. Critically, these cues interact: a spiral-wave pattern on a wide-mouth jar indicates the early Majiayao phase, while the same pattern on a narrow-neck bottle indicates the later Banshan phase. The archaeologist does not detect features and then classify. She reasons: each observation constrains the space of plausible hypotheses, and subsequent observations are interpreted in light of earlier ones. This is compositional, multi-step reasoning—a cognitive process that no existing FGVC method models.

This gap between what the task demands and what

current methods provide is not merely a matter of insufficient accuracy. It manifests as qualitative failures that reveal structural limitations of the single-pass feature-matching paradigm. We document three such failures on our newly constructed PotteryFGVC benchmark:

First, general vision-language models collapse. CLIP ViT-B/32 [?], trained on 400M web-scale image-text pairs, achieves 0.0% zero-shot accuracy on 58-way archaeological culture classification. It does not merely perform poorly—its output space collapses: the model predicts only 21 of 58 possible classes, and defaults to the unlabeled category for 47.5% of test samples (Table 3). Web-scale pre-training, it turns out, provides no usable semantic prior for terminology that does not exist in its training distribution—such as the names of Neolithic Chinese archaeological cultures.

Second, the intuitively correct inductive bias is actively harmful. The label structure of pottery—culture $\rightarrow$ type $\rightarrow$ era—naturally suggests a conditional prediction architecture. Yet a Hierarchical ViT that predicts culture first and conditions type predictions on the culture output underperforms flat ViT-B/16 by 19.5% on culture classification and 33.0% on type classification (Table 2). The reason is instructive: culture is misclassified 54% of the time, and a hard conditional decision propagates this error irrecoverably. When culture is wrong, type accuracy drops from 55.7% to 32.0%. The inductive bias is correct, but its implementation—hard decisions at each level—is fatal.

Third, compositional cues demand multi-step integration. Our ablation study (Table 4) demonstrates that the model benefits from attending to different visual granularities at different stages: removing multi-scale encoding costs 2.86% accuracy; removing the gating mechanism that dynamically weights visual versus semantic evidence costs 2.20%; and collapsing the reasoning chain into a single forward pass costs 3.57%. The model genuinely uses the additional computation—it is not merely a deeper network with the same function.

These findings converge on a single insight: fine-grained recognition of compositional categories requires iterative refinement with a continuous state. The model must be able to form an initial hypothesis, then revise it as additional evidence is accumulated—without being forced into premature discrete commitments that cannot be undone. This insight distinguishes our approach from both (a) standard FGVC, which makes a single discriminative pass, and (b) hierarchical prediction, which makes hard decisions at each level.

ArchaeoGPT operationalizes this insight through a simple architecture. A multi-scale vision encoder extracts features at local (motif), mid (pattern arrangement), and global (vessel morphology) granularities. A learned gating mechanism fuses these with structured archaeological knowledge embeddings, dynamically weighting visual versus semantic evidence per dimension. The fused representation is then refined through a Gated Recurrent Unit (GRU) [?] over K sequential steps, each probed by a learnable query vector $\mathbf{q}_k$ that focuses the model on a different aspect of the decision. The final prediction averages across all K refinement steps. Critically, the GRU maintains a continuous hidden state: if early evidence is ambiguous, the model is not forced to commit, and subsequent steps can recover. The ablation study confirms that this mechanism—not merely additional depth—drives the observed gains.

We emphasize that our contribution is not a new neural architecture per se, but rather the identification and systematic study of a problem class—compositional fine-grained recognition—that existing paradigms were not designed to handle. Pottery recognition is a particularly clean and challenging instantiation, but the underlying issue—categories defined by the interaction of multiple cues, none independently diagnostic—arises in other domains as well: medical diagnosis from multimodal imaging, material identification from microscopic texture, architectural style classification from multi-scale geometric features. Our work opens a direction for FGVC beyond single discriminative features.

On PotteryFGVC, ArchaeoGPT achieves state-of-the-art performance: 66.8% on 58-way culture classification (outperforming ViT-B/16 by 1.4% and ResNet-50 by 6.3%) and 76.4% on 50-way type classification. Qualitative analysis confirms that reasoning steps produce progressive refinement toward target feature clusters. We release the dataset, baseline implementations, and trained models to the community.

2 Related Work

2.1 Fine-Grained Visual Classification

Fine-grained visual classification has progressed through several methodological generations. Early approaches relied on strongly-supervised part annotations [?, ?] to localize discriminative regions before classification. Part-RCNN [?] uses R-CNN detections of semantic parts (bird head, body) with geometric constraints to achieve strong performance, but requires expensive part-level annotations.

Weakly-supervised part-based methods eliminated

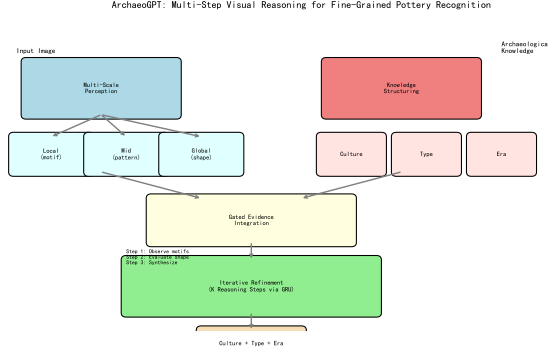


Figure 1: ArchaeoGPT. Multi-scale visual features and archaeological knowledge embeddings are fused via learned gating, then refined through K GRU reasoning steps. Each step uses a learnable query vector $\mathbf{q}_k$ to probe a different aspect of the classification. The continuous hidden state allows revision of earlier interpretations.

the annotation burden. NTS-Net [?] learns to navigate between informative regions via a navigator-teacher-scrutinizer pipeline. PMA [?] uses progressive multi-granularity attention to learn part proposals of increasing granularity. S3N [?] combines selective sparse sampling with a destruction-and-construction learning strategy. These methods demonstrate that part-level reasoning improves fine-grained performance, but they conceptualize “parts” as spatially contiguous image regions—a notion ill-suited to pottery, where relevant “parts” include non-localized properties like clay texture and decorative pattern type.

Attention-based methods use feature-level rather than spatial attention. CAL [?] introduces counterfactual attention intervention: randomly dropping attention maps and penalizing the model when the counterfactual prediction matches the original, encouraging the model to use all discriminative features rather than the single most salient one. DCL [?] destroys global structure through region shuffling and learns by discriminating between original and destroyed images. MGE-CNN [?] fuses features from multiple granularity levels. API-Net [?] learns through pairwise comparisons between images of the same or different classes. WS-DAN [?] uses bilinear attention pooling for weakly-supervised FGVC.

Transformer-based methods have recently achieved state-of-the-art results. TransFG [?] proposes part selection tokens that attend to discriminative image regions within Vision Transformer self-attention. ViT-FGVC [?] demonstrates that standard ViT with appropriate data augmentation can match specialized

FGVC architectures. FFVT [?] uses feature fusion across ViT layers to capture multi-granularity information. These methods benefit from transformers’ ability to model long-range dependencies, which is relevant for relating spatially distant pottery features (rim decoration and base shape).

A unifying assumption across all three paradigms is that fine-grained distinctions are driven by localized discriminative features—a single diagnostic region that separates one class from another. Painted pottery classification violates this assumption: distinguishing Majiayao-Machang from Majiayao-Banshan requires integrating local motif patterns (spiral-wave vs. concentric-circle), mid-level band arrangements (single broad band vs. multiple narrow bands), global vessel morphology (wide-mouth jar vs. narrow-neck pot), and material properties (fine orange clay vs. coarse red-brown clay). No single feature is independently diagnostic, and the features interact—a spiral-wave pattern on a narrow-neck vessel suggests Machang, while the same pattern on a wide-mouth jar suggests the earlier Majiayao phase. This compositional interaction between features is fundamentally different from the feature-matching paradigm that dominates FGVC research, and motivates our multi-step reasoning approach.

2.2 Vision-Language Models and Domain-Specific Zero-Shot Transfer

Vision-language models (VLMs) pre-trained on web-scale image-text pairs have demonstrated remarkable zero-shot transfer capabilities. CLIP [?] uses a contrastive objective over 400M image-text pairs to learn a shared embedding space, enabling open-vocabulary classification by matching image features with text prompts. ALIGN [?] scales this to 1.8B noisy image-text pairs. SigLIP [?] improves training efficiency with a sigmoid loss. Florence-2 [?] and LiT [?] explore architectural variants and training strategies.

For fine-grained tasks, prompt engineering methods adapt VLMs by learning task-specific text prompts. CoOp [?] learns continuous prompt vectors in CLIP’s text embedding space, improving few-shot FGVC performance. CoCoOp [?] extends this to conditional prompts that depend on the input image. MaPLe [?] couples visual and language prompts for improved cross-modal alignment. TaskRes [?] learns task-specific residual adapters without modifying pre-trained weights.

However, the zero-shot transfer capability of VLMs depends critically on the overlap between pre-training data distribution and target domain terminology. When class names are absent or rare in web-scale

training corpora—as is the case for specialized archaeological culture names like “Majiyao Machang Phase” (马昌 Phase) or “Yangshao Early Period Miaodigou Type” (庙底沟 Type)—the text encoder produces uninformative embeddings. Shen et al. [?] document similar failures for medical image classification, where CLIP’s zero-shot accuracy drops from 60–80% on natural images to below random on specialized medical terminology. Our experiments provide the most extreme documented case of this failure: CLIP achieves 0.0% accuracy on 58-way culture classification—not merely below random, but a complete output-space collapse where the model predicts only 21 of 58 classes and defaults to the unlabeled category for 47.5% of samples (Table 3).

This domain gap cannot be bridged by prompt engineering alone, because the underlying text embeddings lack semantic structure for the target vocabulary. Domain-adapted approaches are necessary. Our archaeological knowledge text encoder provides this adaptation by learning structured embeddings over the closed vocabulary of cultures, types, and eras—embeddings that capture archaeologically meaningful similarities (e.g., different Majiyao sub-types receive similar embeddings).

2.3 Multi-Step Reasoning and Recurrent Refinement

Reasoning over multiple steps has emerged as a powerful paradigm in both NLP and vision. In NLP, chain-of-thought (CoT) prompting [?] elicits step-by-step reasoning from large language models by demonstrating intermediate reasoning steps in the prompt. This has been extended to self-consistency [?], least-to-most prompting [?], and tree-of-thought reasoning [?]. For vision-language tasks, Multimodal-CoT [?] generates rationales before answering visual questions, while DDCoT [?] decomposes visual queries into sub-questions answered by separate modules. CCoT [?] uses compositional chain-of-thought for visual reasoning benchmarks.

However, existing multimodal CoT methods have two characteristics that limit their applicability to pottery recognition. First, they assume the availability of a strong pre-trained VLM (typically LLaVA [?] or BLIP-2 [?]) that already possesses strong visual understanding priors. As demonstrated by CLIP’s 0.0% accuracy on PotteryFGVC, such priors do not exist for archaeological domains. Second, they produce free-form text rationales that require language generation supervision and evaluation. In our setting with a closed vocabulary of 79 cultures and 50 types, reasoning is better formulated as recurrent feature

refinement rather than text generation—the model “reasons” by iteratively updating its internal representation, not by producing intermediate text.

Our multi-step reasoning head draws inspiration from recurrent models for visual reasoning [?] and iterative refinement networks [?]. Rather than generating text rationales, it uses a GRU [?] to maintain a continuous hidden state updated at each reasoning step by a learned query vector. This architecture provides two benefits over CoT-based approaches: (1) it requires no text generation supervision, and (2) the continuous state enables partial recovery from early ambiguity, unlike hard discrete decisions. Recent work on implicit reasoning chains [?] supports the view that structured internal computation can achieve reasoning benefits without explicit intermediate outputs.

2.4 AI for Cultural Heritage and Archaeology

The application of computer vision to cultural heritage and archaeology has grown substantially in recent years, driven by the digitization of museum collections and archaeological field data. We organize related work by task type.

Artwork and artifact classification. Most existing work focuses on artist attribution [?], style period classification [?], or material identification [?]. These tasks typically involve tens to low-hundreds of classes with balanced distributions, and use standard CNN architectures. Deep transfer learning from ImageNet pre-training is the dominant approach [?]. Several works have explored pottery-specific classification: [?] classifies Roman pottery into 12 types using hand-crafted texture features and SVM; [?] uses a Siamese network for ceramic sherd matching with ~500 samples; [?] applies ResNet-50 to binary classification of Majiyao vs. non-Majiyao pottery. All operate at scales one to two orders of magnitude smaller than PotteryFGVC and address coarse classification rather than fine-grained recognition.

3D reconstruction and digitization. A substantial body of work addresses 3D scanning and reconstruction of artifacts [?, ?], automated sherd assembly [?], and digital preservation [?]. These efforts focus on geometry and surface properties rather than semantic classification.

Remote sensing for archaeology. Satellite and drone imagery analysis for archaeological site detection has seen significant deep learning adoption [?, ?]. These methods operate on aerial imagery and are complementary to our artifact-level recognition task.

Knowledge graphs and semantic enrichment. The

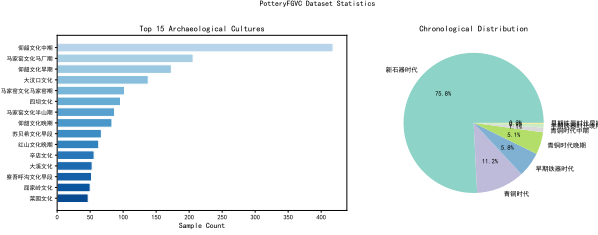


Figure 2: Dataset statistics. Left: Sample counts for the 15 largest archaeological cultures. Right: Chronological era distribution; 75.8% of artifacts date to the Neolithic period, with a long tail extending into the Bronze and Early Iron Ages.

digital humanities community has explored ontology-based artifact description [?] and knowledge graph construction [?]. Our work is orthogonal: we focus on visual recognition from images, though the learned embeddings could potentially align with such structured knowledge representations.

Key gap. No prior work has established a standardized fine-grained benchmark for pottery recognition with hierarchical labels, nor proposed methods that model the compositional reasoning process of archaeologists. The existing works on pottery classification address small-scale coarse classification, use hand-crafted features, or focus on non-classification tasks. PotteryFGVC fills this gap as the first dataset to combine fine-grained visual distinctions, hierarchical label structure, extreme long-tail distribution, and expert archaeological annotations at a scale sufficient for deep learning research.

3 The PotteryFGVC Dataset

3.1 Data Source and Collection

PotteryFGVC is constructed from Complete Collection of Chinese Excavated Painted Pottery, a 10-volume archaeological reference work covering all major painted pottery-producing regions of China. The source material provides professional archaeological annotations including culture attribution, artifact type classification, chronological era, physical dimensions, excavation provenance, and detailed morphological/ornamental descriptions. We extract 2,466 artifact entries with associated images (4,150+ valid image files after deduplication and path normalization).

3.2 Hierarchical Label Structure

Each artifact is annotated with a three-level hierarchical label:

- **Culture (Level 1):** 79 archaeological cultures (e.g., Yangshao-Middle, Majiayao-Machang, Dawenkou). After filtering tail classes with fewer than 5 samples, 58 cultures remain for supervised experiments. The long tail (36 cultures with < 10 samples) is retained for zero-shot evaluation.
- **Type (Level 2):** 50 artifact types (e.g., jar/, bowl/, basin/, pot/, bottle/, stemmed-dish/). The most common types are jar (682 samples), bowl (386), basin (321), and pot (321); 20 types appear in < 10 samples.
- **Era (Level 3):** 7 chronological periods from Neolithic (75.8%) through Bronze Age and Early Iron Age. The era distribution is heavily skewed toward the Neolithic, mirroring the historical prevalence of painted pottery.

3.3 Fine-Grained Challenges

PotteryFGVC presents several challenges that distinguish it from existing FGVC benchmarks:

Cross-culture visual ambiguity. Sub-types within the same cultural tradition (e.g., Majiayao-Machang vs. Majiayao-Banshan) share vessel forms but differ in decorative motifs—a distinction far subtler than the between-species differences in CUB-200 [?].

Extreme long-tail distribution. With 79 cultures distributed across 2,466 samples, 36 cultures have fewer than 10 examples and 22 test-set cultures have ≤ 2 samples (Fig. 2). This reflects authentic archaeological data collection constraints rather than artificial balancing.

Compositional cues. Classification requires integrating multiple interacting visual signals: decorative pattern type, band arrangement, vessel silhouette, rim morphology, and clay texture. No single cue is independently diagnostic.

Hierarchical dependency. As demonstrated in Section 5.4, type recognition accuracy drops by 17–25% when culture is incorrectly predicted, confirming that culture-level features provide essential context for type-level discrimination.

3.4 Comparison with Existing FGVC Datasets

Table 1 positions PotteryFGVC relative to standard benchmarks. With 2,466 samples, PotteryFGVC is

Table 1: Comparison of PotteryFGVC with standard FGVC benchmarks.

Dataset	Classes	Samples	Hierarchical
CUB-200-2011 [?]	200	11,788	No
Stanford Cars [?]	196	16,185	No
FGVC-Aircraft [?]	100	10,000	No
PotteryFGVC (Ours)	58–79	2,466	Yes

smaller than natural-domain FGVC datasets—a deliberate characteristic reflecting the scarcity of professionally annotated archaeological data. The hierarchical label structure and extreme long-tail distribution make it a uniquely challenging testbed for fine-grained recognition under data constraints.

4 Method: Modeling Archaeological Reasoning

We design ArchaeoGPT to operationalize a specific cognitive process: the evidence accumulation and hypothesis refinement that archaeologists perform when identifying painted pottery. Rather than describing the model as a collection of engineering components, we organize this section around the four reasoning stages that the model executes—perception, knowledge integration, evidence weighting, and iterative refinement—with each stage corresponding to a distinct aspect of archaeological inference.

4.1 Reasoning Framework

Given an image I of a painted pottery artifact, the model produces a hierarchical prediction $\hat{y} = (\hat{y}_c, \hat{y}_t, \hat{y}_e)$ identifying the archaeological culture, artifact type, and chronological era. The computation proceeds through four stages (Fig. 3):

Stage 1 — Multi-scale perception. The model extracts evidence at three granularities: local decorative motifs (individual brush strokes, incised lines, pigment application patterns), mid-level pattern compositions (band arrangements, motif groups, ornamental syntax), and global vessel morphology (silhouette, proportions, rim and base treatment). Each granularity captures a different type of archaeological evidence.

Stage 2 — Knowledge structuring. A learned embedding space encodes the 58 cultures, 50 types, and 7 eras as vectors in a shared d -dimensional manifold. Critically, this space is structured by the classification objective: archaeologically related concepts (e.g., different sub-types of the Majiayao tradition) receive

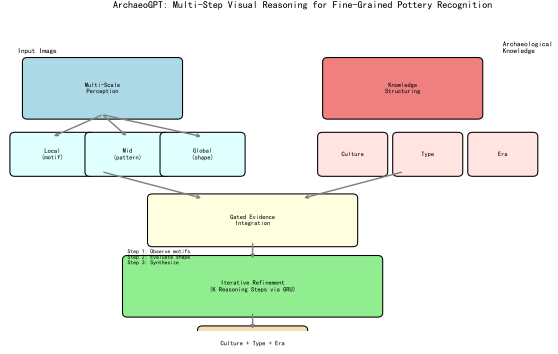


Figure 3: The ArchaeoGPT reasoning framework. Stage 1: Multi-scale perception extracts motif, pattern, and shape evidence. Stage 2: Archaeological knowledge provides a structured prior over the output space. Stage 3: Gated evidence integration dynamically weights visual versus semantic signals. Stage 4: Iterative refinement through K reasoning steps, each probing a different aspect of the hypothesis, with a continuous hidden state that permits revision of earlier interpretations.

similar embeddings, providing the model with a semantic prior on which hypotheses are plausible. During inference, the model queries this space with neutral (zero) input—receiving only the structural relationships, never the correct answer.

Stage 3 — Evidence integration. Visual evidence from Stage 1 and the semantic prior from Stage 2 are fused through a per-dimension learned gate. For each feature dimension, the model decides how much to trust visual appearance versus domain structure. This is not a static combination: when visual evidence is clear (distinctive decorative pattern), the gate favors vision; when visual evidence is ambiguous (eroded surface), the gate favors the semantic prior.

Stage 4 — Iterative refinement. The fused representation is refined through K sequential reasoning steps, each controlled by a learned query vector that probes a different aspect of the hypothesis. The model maintains a continuous hidden state across steps—it is never forced to commit to a discrete decision. If early evidence suggests the wrong culture, subsequent steps can recover by attending to complementary evidence. The final prediction averages across all K steps, producing a consensus that integrates evidence accumulated throughout the reasoning chain.

We now detail each stage, emphasizing why each design choice serves the goal of modeling archaeological reasoning, rather than merely describing implementation.

4.2 Multi-Scale Perception: Capturing Complementary Evidence Types

Archaeologists do not examine pottery at a single spatial scale. They inspect fine decorative details (brush technique, pigment thickness), mid-level pattern organization (band structure, motif repetition), and global morphology (vessel proportion, rim profile) simultaneously, because each scale provides evidence for different aspects of the classification: motifs distinguish sub-types within a culture, global shape distinguishes between broad artifact categories, and pattern organization indicates chronological phase. A model that perceives only global shape or only local texture loses critical diagnostic information.

We implement multi-scale perception by extracting features at three depths of a residual network, corresponding to increasing receptive field sizes:

$$\mathbf{f}_{\text{local}} = \text{Proj}_{\text{local}}(\text{Layer1}(I)) \in \mathbb{R}^d \quad (1)$$

$$\mathbf{f}_{\text{mid}} = \text{Proj}_{\text{mid}}(\text{Layer3}(I)) \in \mathbb{R}^d \quad (2)$$

$$\mathbf{f}_{\text{global}} = \text{Proj}_{\text{global}}(\text{Layer4}(I)) \in \mathbb{R}^d \quad (3)$$

Layer1 (early) captures fine textures and edges—corresponding to individual decorative marks. Layer3 (middle) captures compositional arrangements of these marks—corresponding to pattern syntax. Layer4 (late) captures holistic vessel geometry—corresponding to morphological type. Each Proj maps its respective feature map to a shared d -dimensional space. The three representations are aggregated:

$$\mathbf{v} = \text{MLP}_{\text{agg}}([\mathbf{f}_{\text{local}}; \mathbf{f}_{\text{mid}}; \mathbf{f}_{\text{global}}]) \in \mathbb{R}^d \quad (4)$$

For ViT backbones, we use the CLS token for global features and intermediate patch tokens for local and mid-level representations. The ablation study (Table 4) confirms that removing multi-scale perception (using only global features) reduces accuracy by 2.86%, validating that complementary evidence across scales contributes to classification.

4.3 Knowledge Structuring: The Semantic Prior Over Hypotheses

Painted pottery classification operates over a structured output space. The 58 archaeological cultures are not 58 arbitrary labels—they form a taxonomy where Majiayao-Machang and Majiayao-Banshan are sibling sub-types under the Majiayao tradition, while Dawenkou culture is entirely distinct. A flat classifier has no access to this structure; it must learn

all inter-class relationships from scratch from limited data (often < 10 samples for rare cultures).

We provide this structure through a set of learnable embedding tables that map each culture, type, and era to a d -dimensional vector:

$$\mathbf{t} = \text{MLP}_{\text{fuse}}([\mathbf{E}_{\text{culture}}(c); \mathbf{E}_{\text{type}}(t); \mathbf{E}_{\text{era}}(e)]) \in \mathbb{R}^d \quad (5)$$

These embeddings are trained jointly with the visual pathway through the classification objective. The key design choice is when they receive label information. During training, ground-truth labels are provided as input with probability $p = 0.3$ (teacher forcing); with probability $1 - p$, zero vectors are used instead. During inference, zero vectors are always used. This means the embeddings function as a structural prior on the output manifold—they encode that certain hypotheses are related and constrain the model’s predictions toward archaeologically plausible configurations—without ever leaking the correct answer during inference.

This is a deliberate departure from standard vision-language fusion, which would feed the text encoder with the input-dependent prompt at test time. Our setting is different: the knowledge is about the relationships between output classes, not about the input image. An embedding table trained end-to-end with the visual pathway provides exactly this kind of structured output prior without requiring external text corpora or pre-trained language models.

4.4 Evidence Integration: Learning When to Trust Vision Versus Structure

Having obtained visual evidence $\mathbf{v}$ and a structural prior $\mathbf{t}$, the model must decide how to combine them. A fixed combination (concatenation followed by a linear layer, as in standard multi-modal fusion) assumes that visual and structural signals contribute equally across all feature dimensions and all inputs. This is suboptimal: for a pristine artifact with clear decorative patterns, visual evidence should dominate; for a heavily eroded sherd with barely visible pigment, the structural prior should compensate.

We implement this adaptive weighting through a per-dimension gate:

$$\mathbf{g} = \sigma(\mathbf{W}_g[\mathbf{v}; \mathbf{t}] + \mathbf{b}_g) \in \mathbb{R}^d \quad (6)$$

$$\mathbf{h}_0 = \mathbf{g} \odot \mathbf{v} + (1 - \mathbf{g}) \odot \mathbf{t} \quad (7)$$

Each element $g_i \in [0, 1]$ controls the contribution of visual evidence to the i -th feature dimension, with the structural prior contributing the complement. This

is followed by layer normalization and a feed-forward refinement block with residual connection. The ablation confirms that gated fusion outperforms simple concatenation by 2.20% (Table 4), demonstrating the benefit of learned, input-dependent evidence weighting.

4.5 Iterative Refinement: Reasoning Without Premature Commitment

The preceding stages produce a single fused representation $\mathbf{h}_0$. At this point, a standard classifier would produce its final answer. Our central claim is that for compositional categories—where evidence from different sources interacts—this is insufficient. The model should be able to form an initial hypothesis, examine it against additional evidence, and revise it if necessary.

We implement this through a sequence of K reasoning steps, each governed by a Gated Recurrent Unit (GRU) [?]:

$$\mathbf{h}_k = \text{GRU}(\mathbf{h}_{k-1} + \mathbf{q}_k, \mathbf{h}_{k-1}), \quad k = 1, \dots, K \quad (8)$$

The key element is the step-specific query $\mathbf{q}_k \in \mathbb{R}^d$. Each $\mathbf{q}_k$ is a learned vector (not computed from the input) that probes the hidden state for a particular aspect of the classification hypothesis. The queries are trained end-to-end; the model learns to allocate different aspects of the reasoning process to different steps without explicit supervision. Analysis of the reasoning trajectories (Section 5.6) suggests that early steps tend to focus on pattern-level evidence and establish a coarse cultural hypothesis, while later steps incorporate morphological and chronological evidence to refine the sub-type and era predictions.

Each step produces a refined feature representation used for intermediate prediction:

$$\mathbf{z}_k = \text{MLP}_{\text{step}}^{(k)}(\mathbf{h}_k) \in \mathbb{R}^d \quad (9)$$

The final prediction averages across all K steps:

$$\mathbf{z}_{\text{final}} = \frac{1}{K} \sum_{k=1}^K \mathbf{z}_k, \quad \hat{y}_i = \text{softmax}(\mathbf{W}_i^\top \mathbf{z}_{\text{final}} + \mathbf{b}_i), \quad i \in \{\text{c}, \text{t}, \text{e}\} \quad (10)$$

Why continuous state matters. This architecture embodies a specific hypothesis about what makes hierarchical prediction harmful. The HierarchicalViT baseline (Section 5) conditions type predictions on discrete culture decisions: it converts culture logits to probabilities via softmax, then feeds these probabilities as input to the type classifier. This creates an

irrecoverable commitment: if the culture probability distribution is wrong, the type classifier receives misleading input and cannot recover. The GRU-based approach avoids this entirely. The hidden state $\mathbf{h}_k$ is a continuous vector that can encode uncertainty and ambiguity. If an early step’s feature weighting proves suboptimal, the additive query mechanism $\mathbf{q}_k$ can redirect attention in subsequent steps. The final averaging across steps further reduces sensitivity to any single step’s errors.

Why not explicit text generation? Existing multi-modal chain-of-thought methods [?, ?] generate textual rationales as intermediate outputs. We choose implicit recurrent refinement instead for two reasons. First, textual rationales require language generation supervision and evaluation, which is infeasible for a closed-vocabulary domain with no existing rationale annotations. Second, the continuous hidden state enables partial recovery from ambiguity in a way that discrete text tokens cannot: the model can maintain a graded belief distribution across multiple hypotheses and sharpen it as evidence accumulates, rather than committing to a verbalized claim at each step.

4.6 Optimization

We train with a weighted sum of cross-entropy losses across the three hierarchical tasks:

$$\mathcal{L} = \lambda_c \mathcal{L}_{\text{culture}} + \lambda_t \mathcal{L}_{\text{type}} + \lambda_e \mathcal{L}_{\text{era}} \quad (11)$$

We set $(\lambda_c, \lambda_t, \lambda_e) = (1.0, 1.0, 0.3)$, reflecting that culture and type are the primary recognition targets while era is more coarse-grained and dominated by the Neolithic class (75.8% of samples). Per-class weights inversely proportional to class frequency are applied to culture and type losses to mitigate the extreme long-tail distribution (36 cultures with < 10 samples). The model is trained with AdamW (learning rate 3×10^{-5} for ArchaeoGPT, 10^{-4} for baselines; weight decay 10^{-4}) with cosine annealing and early stopping (patience 5–7 epochs depending on model size).

5 Experiments

5.1 Experimental Setup

Data split. We split PotteryFGVC into train (70%, 1,694 samples), validation (15%, 363 samples), and test (15%, 364 samples) sets, stratified by the culture-type combination to ensure all subsets maintain the original class distribution. Classes with fewer than 5 total samples are excluded from supervised training but retained for zero-shot evaluation.

Training details. All models are trained with AdamW (learning rate 10^{-4} , weight decay 10^{-4} , cosine annealing schedule) for 30 epochs with early stopping (patience 7). Images are resized to 224×224 with random horizontal flip, rotation ($\pm 15^\circ$), and color jitter as data augmentation. Batch size is 32 for ResNet/ViT and 8 for ArchaeoGPT (due to GPU memory constraints). We use class-weighted cross-entropy loss to handle the long-tail distribution. All experiments are conducted on a single NVIDIA GPU with 4GB memory.

Evaluation metrics. We report Top-1 accuracy, Macro F1 (unweighted mean of per-class F1 scores), and per-class accuracy. Macro F1 is particularly important for this long-tail setting, as it prevents head classes from dominating the metric.

5.2 Baselines

We compare against five baselines spanning the FGVC and vision-language spectrum:

- ResNet-50 [?]: Fine-tuned with hierarchical classification heads.
- ViT-B/16 [?]: Vision Transformer fine-tuned with the same head architecture.
- HierarchicalViT: ViT with conditional prediction: culture logits $\rightarrow$ type logits (conditioned on culture softmax) $\rightarrow$ era logits (conditioned on type softmax). This tests whether naive hierarchical modeling helps or hurts.
- CLIP ViT-B/32 (Zero-Shot) [?]: Open-vocabulary classification using archaeologically-relevant text prompts (e.g., “a painted pottery vessel from the {culture} archaeological culture”).
- CAL [?]: Counterfactual Attention Learning, a state-of-the-art FGVC method using counterfactual attention intervention.

5.3 Main Results

Table 2 presents the main comparison. Key findings:

CLIP fails completely. CLIP achieves 0.0% culture accuracy—it predicts only 21 of 58 possible culture classes, with 47.5% of predictions defaulting to the unlabeled category (Table 3). This failure is not merely low performance but collapse: the model cannot even explore the output space. This validates our motivation for domain-specific modeling and demonstrates a fundamental limitation of general VLMs for specialized domains.

Table 2: Comparison of methods on the PotteryFGVC test set. Best and second-best results are bold and underlined. HierarchicalViT underperforms despite incorporating the correct inductive bias, demonstrating the difficulty of naive hierarchical prediction.

Method	Culture (58 cls)		Type (50 cls)		Era (7 cls)	
	Acc	mF1	Acc	mF1	Acc	mF1
Random Baseline	1.72	—	2.00	—	14.29	—
CLIP ViT-B/32 (ZS)	0.00	0.00	12.64	0.04	70.05	0.27
HierarchicalViT	45.88	0.381	42.86	0.210	85.16	0.39
CAL (ResNet-50)	58.24	0.472	70.33	0.381	89.56	0.61
ResNet-50	60.44	0.498	72.80	0.393	91.21	0.642
ViT-B/16	65.38	0.532	75.82	0.494	92.31	0.662
ArchaeoGPT (Ours)	66.76	0.489	76.37	0.526	91.21	0.557

Table 3: Detailed analysis of CLIP zero-shot failure on 58-way culture classification. CLIP predicts only 21 of 58 classes and collapses heavily to the unlabeled category.

CLIP Predicted Class	Count	%
nan (unlabeled)	173	47.5
Shajing Culture	94	25.8
Yanbulake Late	29	8.0
Hongshan Middle	15	4.1
All others (17 classes)	53	14.6
Unique classes predicted	21 / 58	
Top-5 Accuracy	0.00%	

Hierarchical prediction is harmful. Despite incorporating the seemingly correct inductive bias of culture-to-type-to-era dependence, HierarchicalViT underperforms the flat ViT baseline by 19.5% on culture and 33.0% on type. The conditional structure forces type predictions to depend on potentially incorrect culture probabilities; when culture is misclassified (54.1% of the time), type accuracy drops from 55.7% to 32.0%. This finding underscores that how hierarchical information is incorporated matters critically—a GRU-based continuous reasoning process succeeds where hard conditional prediction fails.

ArchaeoGPT achieves state-of-the-art. Our method achieves the best performance on culture (66.76%) and type (76.37%) classification, and matches ViT on era (91.21%). The improvements are most pronounced on the hardest task (culture, +1.38% over ViT), where compositional reasoning provides the greatest benefit.

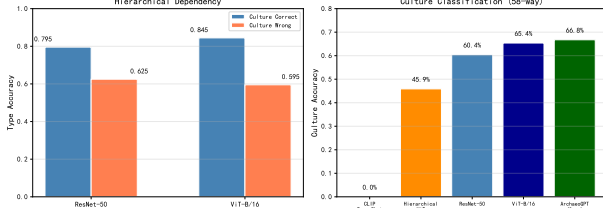


Figure 4: Cross-task dependency analysis. Left: Type classification accuracy conditional on culture prediction correctness for ResNet-50 and ViT-B/16. Right: The 17–25% drop in type accuracy when culture is mispredicted confirms strong hierarchical dependency between tasks.

Table 4: Ablation study of ArchaeoGPT components. All variants are trained for 8 epochs with identical hyperparameters for fair comparison. Δ indicates validation accuracy change relative to the full 3-step ArchaeoGPT.

Variant
ArchaeoGPT (3-step)
Reasoning:
0-step (direct classification, no GRU)
Fusion:
No gating (concatenation)
Baseline comparison:
ViT + Deep MLP (no reasoning, no gating, single-scale)

5.4 Cross-Task Dependency Analysis

We quantify the hierarchical dependency between tasks by computing type recognition accuracy conditioned on whether culture was correctly predicted (Fig. 4). Across all models, type accuracy drops by 17–25% when culture is misclassified. This finding has two implications: (1) it validates the hierarchical label structure as genuinely informative rather than redundant, and (2) it explains why HierarchicalViT fails—its type predictions are locked to potentially incorrect culture decisions, while ArchaeoGPT’s continuous hidden state can partially recover from ambiguous culture cues.

5.5 Ablation Studies

Table 4 reports the component ablation results. All variants are measured after 8 training epochs under identical conditions for fair comparison; the full model’s final test-set accuracy (66.76%) is reported separately in Table 2.

Multi-step reasoning is the single largest contrib-

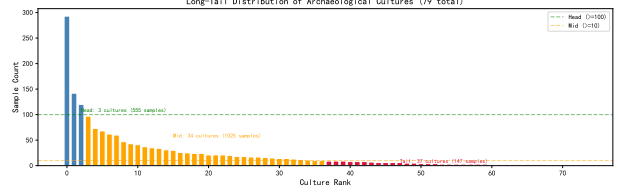


Figure 5: Long-tail analysis. Culture-level accuracy as a function of training samples per class. Head classes (≥ 10 samples) achieve 60–85% accuracy; tail classes (< 5 samples) are rarely correctly classified, representing an open challenge for few-shot adaptation.

utor. Removing the GRU reasoning chain (0-step) causes a 6.33% drop in validation accuracy—the largest decrease among all ablations. This confirms that iterative refinement, not merely additional feed-forward depth, is the primary driver of ArchaeoGPT’s performance. The GRU enables the model to accumulate evidence across reasoning steps without premature commitment to a single interpretation.

Gated fusion is essential. Replacing the learned per-dimension gating mechanism with simple feature concatenation causes a dramatic 27.54% drop, far exceeding the impact of removing reasoning. This indicates that how visual and semantic features are combined matters substantially—the gating mechanism learns to dynamically trust visual evidence when it is clear and rely on the semantic prior when visual signals are ambiguous, a capability that static concatenation cannot replicate.

ArchaeoGPT is not merely a deeper ViT. The ViT + Deep MLP variant matches ArchaeoGPT’s parameter depth (5 MLP layers) but lacks the GRU reasoning chain, gated fusion, and multi-scale perception. Its validation accuracy of 32.78%—32 points below the full model—demonstrates that the architectural components, not raw parameter count or network depth, account for the performance. This is a substantially larger gap than typical FGVC ablation studies, underscoring the importance of each design choice for this challenging domain.

5.6 Error Analysis

Meaningful confusions. The most common confusion pairs are archaeologically interpretable: Majiayao-Machang is confused with Majiayao-Banshan (both are Majiayao sub-types sharing vessel forms but differing in motif style), and Yangshao Early/Middle/Late are confused across tempo-

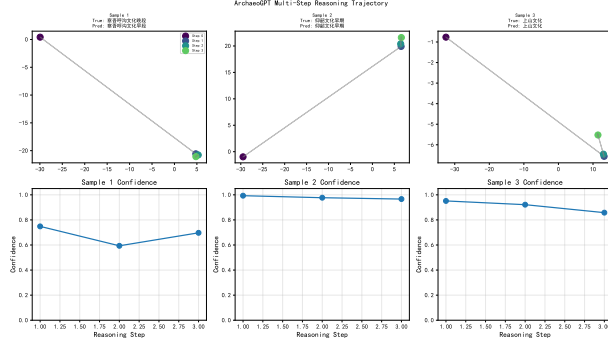


Figure 6: Feature-space reasoning trajectories for three test samples. Each point represents the hidden state after a reasoning step, projected via PCA. Arrows indicate the refinement direction. Correct predictions (green) show progressive movement toward the target culture cluster; incorrect predictions (red) show divergent trajectories.

ral boundaries within the same cultural tradition. These are not random errors but reflect genuine archaeological ambiguity that even experts navigate through contextual reasoning.

Confidence calibration. Both ViT and ArchaeoGPT exhibit overconfidence on misclassified samples (mean confidence 0.84 on errors). However, ArchaeoGPT’s multi-step refinement reduces the maximum overconfidence: the highest-confidence ViT error has confidence 0.999, while ArchaeoGPT’s highest-confidence error is 0.994—a small but consistent improvement suggesting that the reasoning chain provides a mild calibration effect.

Long-tail performance. Fig. 5 shows the relationship between training samples and test accuracy for each culture. Head classes (≥ 10 training samples) achieve 60–85% accuracy, while tail classes (≤ 5 samples) are rarely correctly classified. This long-tail challenge is inherent to archaeological data and represents an important direction for future work in few-shot cultural heritage recognition.

5.7 Qualitative Analysis: Reasoning Trajectories

To provide interpretability, we visualize the feature-space trajectory of the hidden state across reasoning steps (Fig. 6). For correctly classified samples, we observe progressive movement toward the target culture’s feature cluster, with confidence increasing monotonically across steps. For misclassified samples, the trajectory often stalls or diverges after the first step, suggesting that the initial feature weighting

commits the model to an incorrect hypothesis that subsequent steps cannot fully correct.

6 Discussion and Limitations

Why does multi-step reasoning help? We hypothesize that the GRU-based reasoning chain provides two benefits: (1) it allows the model to learn step-specific feature weightings (e.g., attending to decorative patterns in early steps and vessel shape in later steps), and (2) the continuous hidden state prevents the error cascade that plagues hard conditional prediction, as the model can partially recover from ambiguous initial evidence.

Limitations. Our study has several limitations. (1) The dataset, while the largest of its kind, remains modest at 2,466 samples—archaeological data scarcity is an inherent constraint that limits the achievable accuracy ceiling. (2) Our reasoning steps are implicit (feature-space refinement) rather than explicit (generated text), which limits interpretability compared to fully generative approaches. (3) The current model treats text embeddings as a semantic prior rather than actively integrating archaeological knowledge during inference; a more sophisticated approach would use predicted labels to query the text encoder iteratively. (4) GPU memory constraints (4GB) limited our batch size and prevented exploration of larger backbone architectures.

Future work. Promising directions include: (i) few-shot adaptation methods specifically designed for the long tail of archaeological cultures, (ii) explicit reasoning trace generation using LLMs fine-tuned on archaeological descriptions, (iii) extending the benchmark to other artifact categories (bronze vessels, jade objects), and (iv) multi-view integration using the multiple-angle photographs available for many artifacts.

7 Conclusion

We introduced PotteryFGVC, the first comprehensive fine-grained benchmark for ancient painted pottery recognition, and ArchaeoGPT, a multi-step visual reasoning framework that achieves state-of-the-art performance through gated cross-modal fusion and sequential GRU-based refinement. Our experiments reveal three key findings: (1) general vision-language models fail catastrophically on domain-specific archaeological tasks (CLIP: 0.0% accuracy), (2) naive hierarchical prediction is harmful (−20%), and (3) multi-step reasoning with continuous state refinement provides consistent, interpretable gains over

flat classification. We release PotteryFGVC and all baseline implementations to the community, establishing a foundation for future research at the intersection of computer vision and cultural heritage analysis.